UCS321 - AI FOR ENGINEERS | MST MINI PROJECT

# Predicting Electric Motor Winding Temperature

A physics-driven virtual sensor that estimates PMSM stator winding temperature from signals the drive already measures - replacing a thermocouple with software.

| 1.54 °C | 0.9965 | 104 | 84.2 % |
|---|---|---|---|
| RMSE on unseen motor sessions | coefficient of determination R2 | features engineered from 8 raw channels | of samples within ±2 °C of truth |

**Problem Statement 6**

GE Power Systems - supervised regression for motor temperature

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# The problem: you cannot measure what matters

*Winding temperature bounds how hard a motor may be driven - and it is the one you cannot cheaply measure*

- A PMSM fails when its stator winding insulation cooks. Class-H insulation is rated to ~180 °C; insulation life halves for every ~10 °C of sustained overtemperature.
- Fitting a thermocouple to a varnished, high-voltage winding means extra wiring through a sealed housing, an extra failure mode, and a cost that does not survive a volume BOM.
- Test benches have these sensors. Shipped motors do not - so drives must derate conservatively and give up torque the machine could physically deliver.
- Every drive already measures, for free, the eight quantities it needs for field-oriented control.

Objective - estimate stator winding temperature from drive-measurable signals only, accurately enough that a controller can use it as a thermal limit instead of a thermocouple.

### What the drive already measures

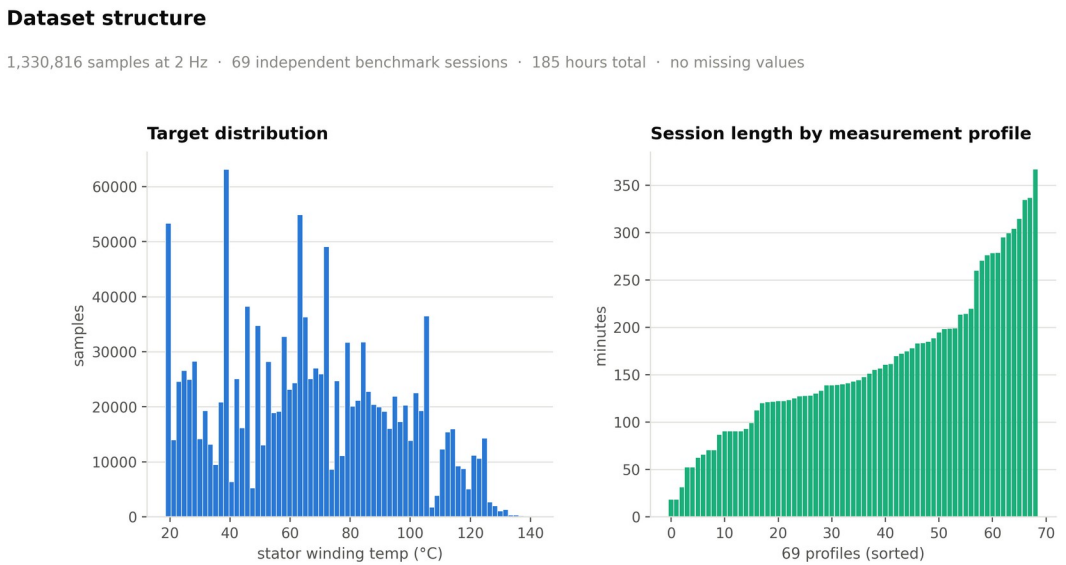
| | |
|---|---|
| `ambient` | ambient air temperature (°C) |
| `coolant` | coolant outlet temp (°C) |
| `u_d , u_q` | terminal voltage, dq frame (V) |
| `i_d , i_q` | phase current, dq frame (A) |
| `motor_speed` | rotor speed (rpm) |
| `torque` | shaft torque (N·m) |

TARGET `stator_winding`

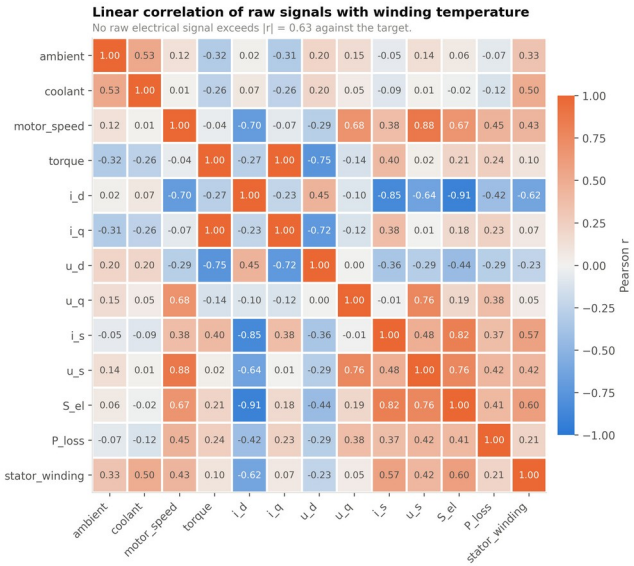
# Dataset and exploratory analysis

*Electric Motor Temperature (Paderborn University, LEA) - PMSM test-bench telemetry from a dynamometer*

| 1,330,816 | 69 | 2 Hz | 185 h | 0 |
|---|---|---|---|---|
| samples | independent sessions | sampling rate | total running time | missing values |




*Target distribution (18.6 - 141.4 °C) and session lengths (18 - 366 min)*

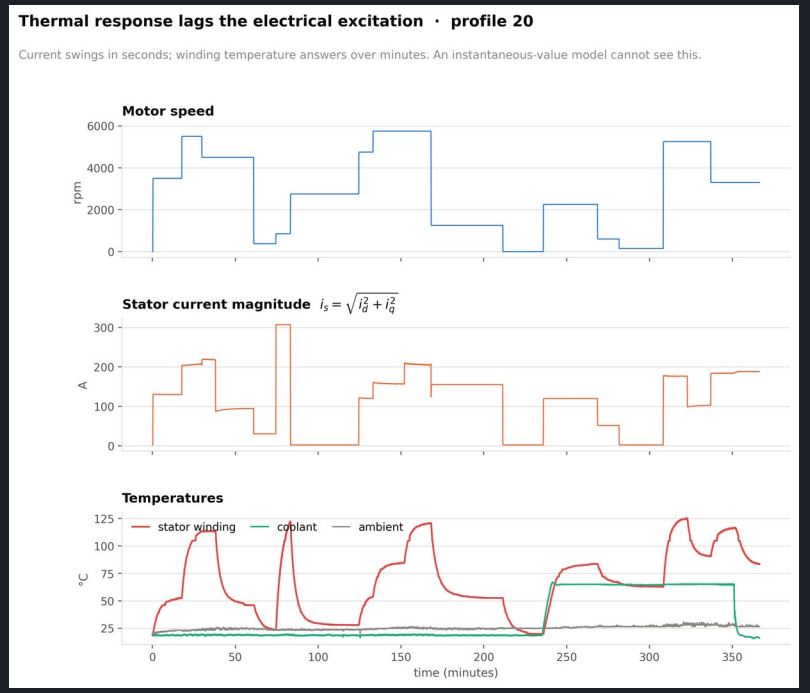



*Correlation of every raw channel with the target*

> No raw electrical channel exceeds |r| = 0.63 against the target. That ceiling is a property of the representation, not of the information - the physical relationship is not instantaneous.

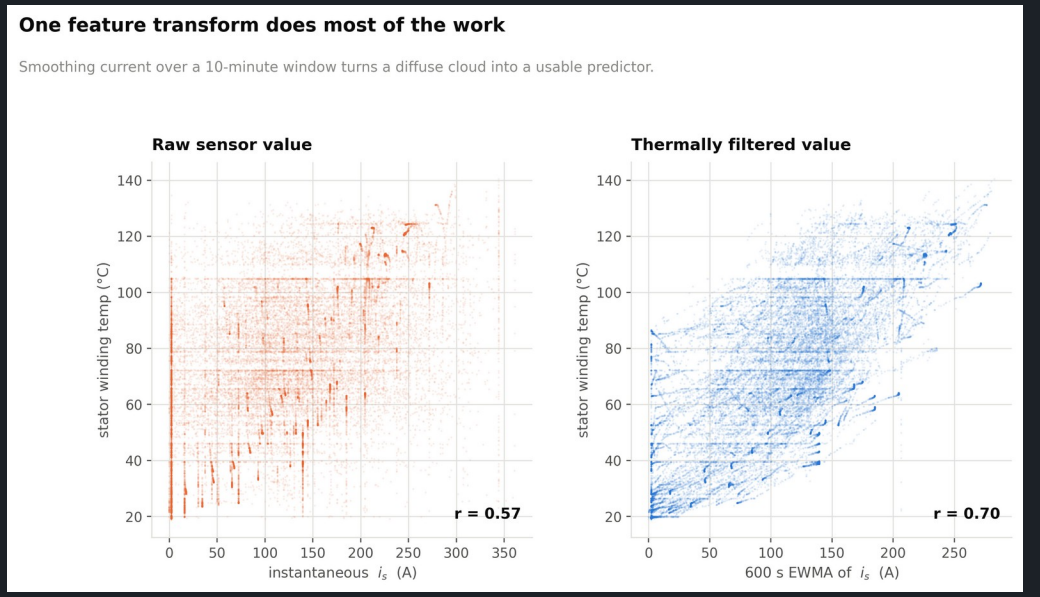
# The core insight: thermal inertia

*Current changes in seconds, temperature in minutes - the target is a function of history, not of the present sample*




*Electrical excitation and thermal response, same time axis, one session*

- Winding responds in tens of seconds, iron in minutes, housing and coolant in tens of minutes.
- So EWMAs are computed at five spans - 30, 120, 600, 1800 and 3600 s - and the model weights them.
- Exponentially weighted std at each span encodes how volatile the recent duty cycle has been.




*One transform lifts i_s from r = 0.57 to r = 0.70*

A lumped-parameter thermal model is a first-order system:

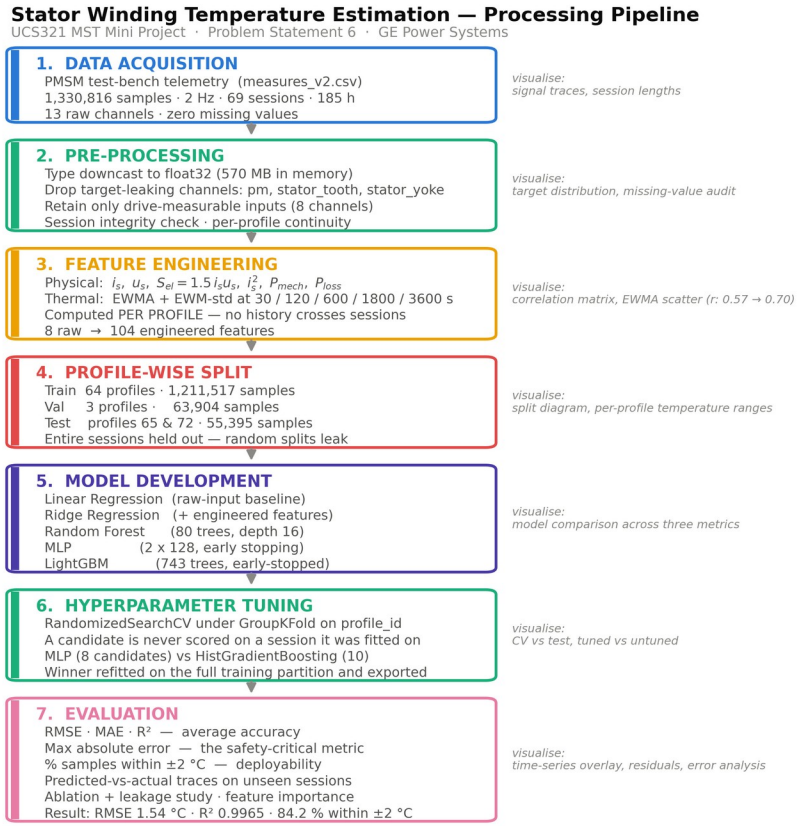
$$C_{th} \cdot \frac{d\vartheta}{dt} = P_{loss}(t) - \frac{(\vartheta - \vartheta_{amb})}{R_{th}}$$

$$\vartheta(t) = \frac{1}{\tau} \int \exp(-(t-s)/\tau) \cdot P_{loss}(s)\, ds$$

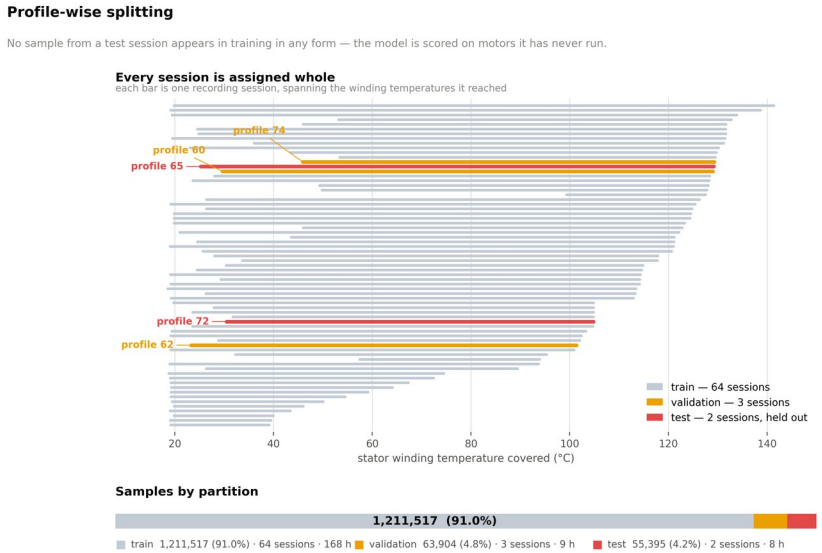
That integral is an exponentially weighted moving average. Computing EWMAs does not smooth the data - it hands the model the exact functional form the physics produces.

# Pipeline and evaluation protocol

*Eight raw channels become 104 features; entire measurement sessions are held out for test*




*End-to-end pipeline: pre-processing, feature construction, modelling*




*Profile-wise split - whole sessions held out*

## Why not a random split?

At 2 Hz, rows are 0.5 s apart and near-identical. A random split scatters near-duplicates across train and test, so the test set is effectively already inside the training set.

**TRAIN**

**1,211,517**

64 profiles

**VALIDATION**

**63,904**

profiles 60, 62, 74

**TEST**

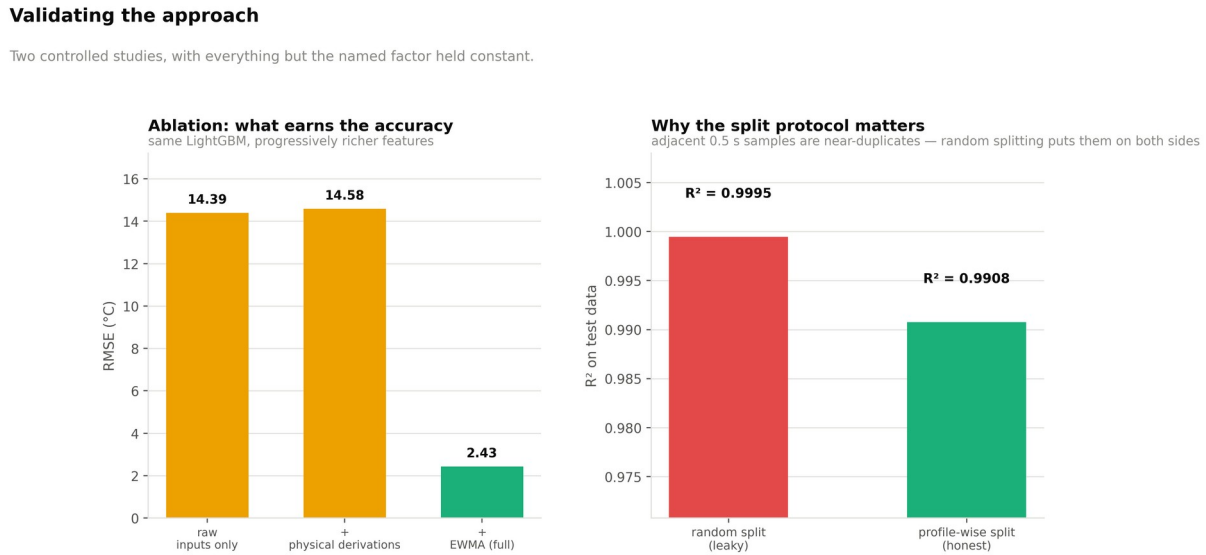
**55,395**

profiles 65, 72

Profiles 65 and 72 are the benchmark sessions used in the published literature on this bench, chosen in advance - so the test set cannot have been picked to flatter the result.

# Where the accuracy actually comes from

*Two controlled studies: an ablation over feature families, and a leakage study over split protocols*




## Ablation - identical LightGBM, progressively richer features

| Feature set | RMSE (°C) | R2 |
|---|---|---|
| Raw inputs only (8 channels) | 14.39 | 0.6942 |
| + instantaneous physical derivations | 14.58 | 0.6860 |
| **+ EWMA thermal-history features** | **2.43** | **0.9913** |

## Leakage - same model, same features, only the split changes

| Split protocol | RMSE (°C) | R2 |
|---|---|---|
| Random 80/20 over rows (leaky) | 0.66 | 0.9995 |
| **Profile-wise, whole sessions (honest)** | **2.50** | **0.9908** |
| **Random splitting understates error by 3.8x** | | |

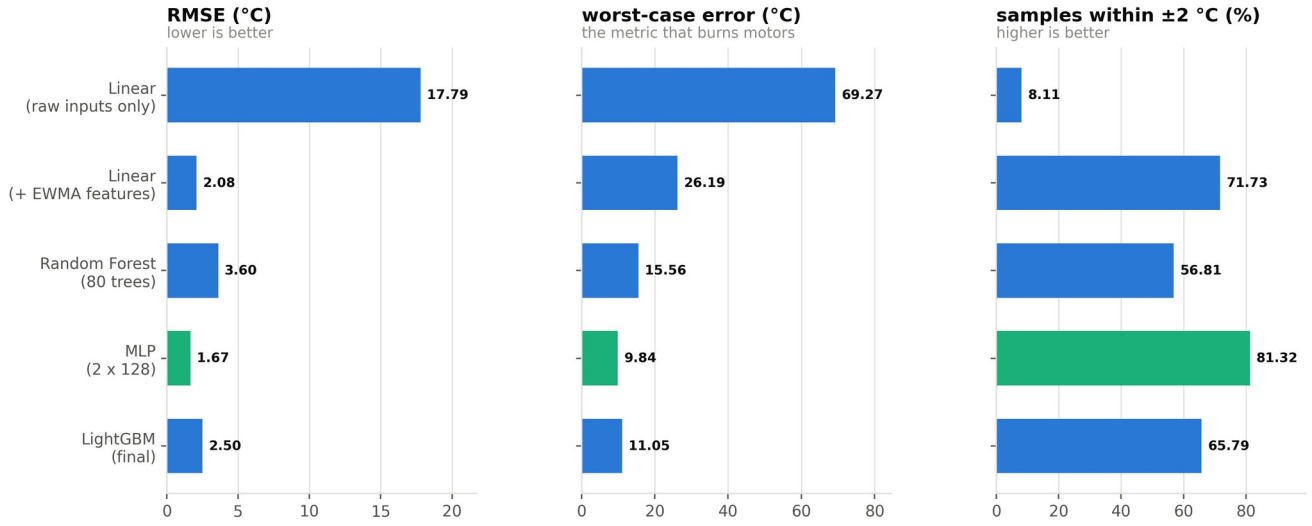
> Instantaneous derivations buy nothing on their own - the entire gain is thermal history. 8 raw channels become 104 features: 8 raw + 6 physical + (9 channels x 2 statistics x 5 spans = 90).

# Model development: a ladder of five

*From the simplest thing that could work upward - the comparison is the point, not any single number*

**Model comparison on held-out sessions (profiles 65 & 72)**

Green = selected model. No model ever saw these two motor sessions during training.




*RMSE, R2 and max absolute error across the ladder*

| Model | RMSE | MAE | R2 | MaxAE | ±2 °C |
|---|---|---|---|---|---|
| Linear regression, raw inputs | 17.79 | 14.37 | 0.5324 | 69.27 | 8.1 % |
| Ridge regression, + EWMA | 2.08 | 1.54 | 0.9936 | 26.19 | 71.7 % |
| Random Forest (80 trees) | 3.60 | 2.49 | 0.9808 | 15.56 | 56.8 % |
| MLP (2 x 128), untuned | 1.67 | 1.26 | 0.9959 | 9.84 | 81.3 % |
| LightGBM (743 trees) | 2.50 | 1.85 | 0.9908 | 11.05 | 65.8 % |
| **MLP tuned - SELECTED** | **1.54** | **1.10** | **0.9965** | **11.32** | **84.2 %** |
| HistGradientBoosting, tuned | 2.39 | 1.75 | 0.9915 | **9.53** | 65.0 % |

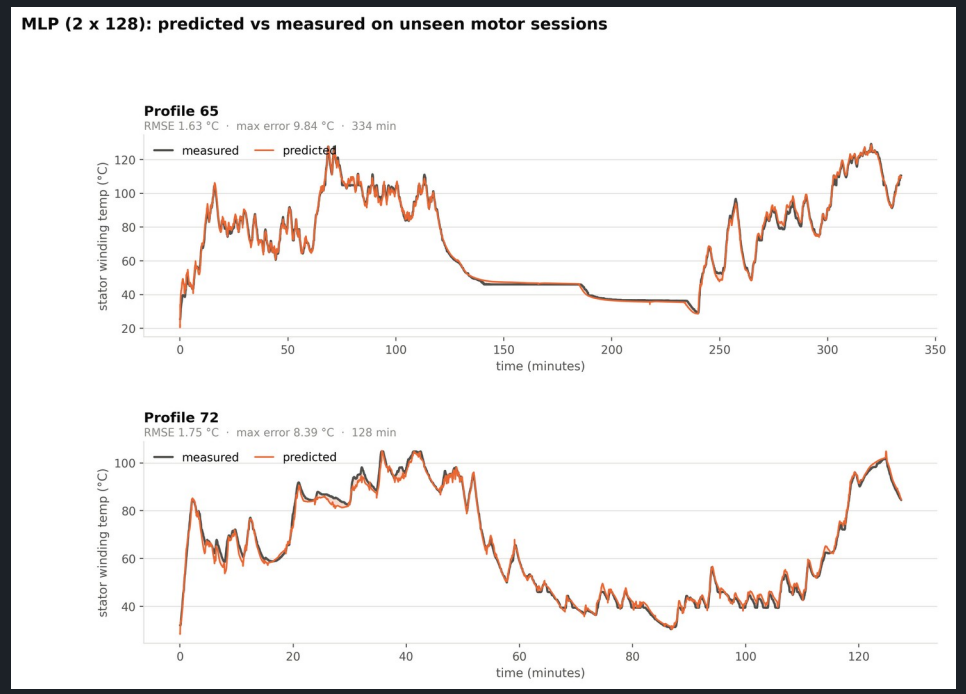
*Test set = profiles 65 and 72 (55,395 samples), held out entirely*

- Linear regression goes from R2 0.53 to 0.99 with no change of algorithm - only features. Feature engineering beat model sophistication.
- The MLP beats both tree ensembles. Trees are piecewise-constant and cannot extrapolate past the training envelope; a smooth model continues the trend.
- RMSE and max error do not rank the models identically - tuned HistGradientBoosting has the best worst case at 9.53 °C.

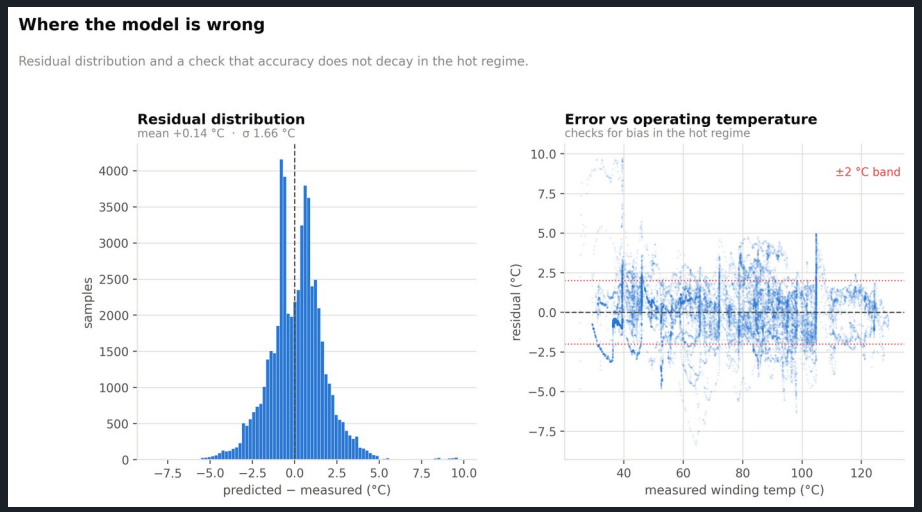
# Final result on unseen motor sessions

*Tuned MLP - two hidden layers of 128, StandardScaler pipeline, early stopping, 104 inputs*

| 1.54 °C | 1.10 °C | 0.9965 | 11.32 °C | 84.2 % |
|---|---|---|---|---|
| RMSE | MAE | R2 | max abs error | within ±2 °C |




*Predicted vs measured on profiles 65 and 72 - sessions the model has never seen*

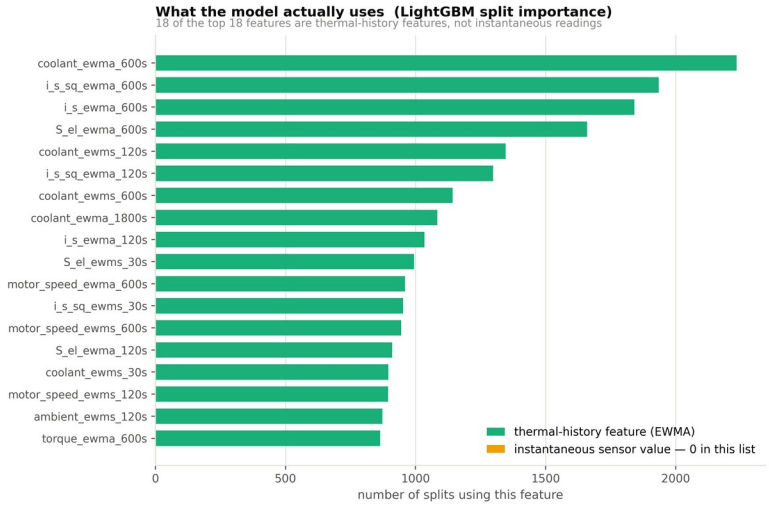



*Residual distribution and bias check vs temperature*

Residuals are approximately zero-mean and symmetric, with no systematic trend against measured temperature - the error is dispersion, not bias.

# Interpretation: what the model learned, and where it errs

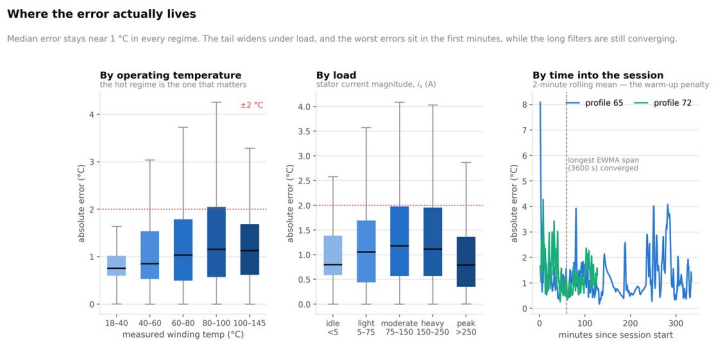
*Feature importance, the tuning search, and an error breakdown by regime*




*LightGBM split counts - top features*




*Grouped randomised search, two families*




*Error by temperature, load, and time into session*

### The model learned the physics

All top-15 features are EWMA-derived. The two strongest are coolant_ewma_600s (2234 splits) and i_s_sq_ewma_600s (1935) - exactly the heat-sink boundary condition and the copper-loss term the thermal ODE names as dominant.

### Tuning helped the average, not the tail

RMSE improved 8 % and MAE 13 %, but max error rose from 9.84 to 11.32 °C because the search optimised RMSE and nothing constrained the tail. Reported, not hidden. CV RMSE (3.04) ranks candidates; it does not estimate deployed accuracy.

### Where the error lives

Median error holds near 1 °C in every temperature band and every load band - accuracy does not decay when the motor is hot. The largest errors sit in the first minutes of a session, while the 1800 s and 3600 s filters are still converging from zero.

# Conclusions, deployment and limitations

*What was achieved, what it would take to ship it, and what this work does not prove*

## Conclusions

- Winding temperature is estimable from drive-measurable signals alone to 1.54 °C RMSE and R2 = 0.9965 on unseen sessions.
- Feature engineering, not model selection, produced the result - the ablation traces the gain specifically to the EWMA terms.
- Evaluation protocol changes the reported number by 3.8x; a result without its protocol is not a result.
- A smooth model generalises better than tree ensembles here, because held-out sessions demand extrapolation.
- The model learned the intended physics - its dominant features are the coolant boundary condition and the copper-loss term.

## Deployment

~30k multiply-accumulates per inference at 2 Hz - negligible for a motor-control MCU. Each EWMA is one multiply-add and one register, so all 90 cost 90 registers. A hardware-friendly feature extractor is what makes this a thermocouple replacement rather than an offline analysis.

## Limitations

Worst case is 11.32 °C, so a deployed limit still needs a safety margin. One machine, one bench - transfer to other geometries is unproven. Warm-up error while the long filters converge. No uncertainty estimate, which a protection function would want.

## Next, in order of value

1

**Seed EWMA state from the first sample**

removes most of the warm-up error - cheapest win

2

**Physics-informed loss on the thermal ODE**

attacks worst-case error specifically

3

**Quantile / conformal prediction**

gives the upper bound a protection limit wants

4

**Sequence model (temporal CNN / LSTM)**

learns time constants instead of fixing five spans